Fiscal Year:	FY 2013	**Task Last Updated:**	FY 07/09/2013
PI Name:	Ray, F. Andrew Ph.D.		
Project Title:	NEXGEN Approaches to Chromatid Painting		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80524	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Crew Health NNJ11ZSA002NA
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No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
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Task Description:	There is every reason to believe that large and small scale chromosomal intra-changes (inversions) are just as important as inter-changes between chromosomes, e.g., translocations, the latter of which are well known to cause cancer. And yet, inversions, especially small ones, have been historically difficult, if not impossible to detect. Recently, KromaTiD Inc., using SBIR funding from NASA, successfully demonstrated a new platform technology to detect small chromosomal inversions using directional oligonucleotide probes called chromatid painting. Here, we propose to develop the next generation of whole chromatid paints using a much faster, more cost effective approach. If successful, this will facilitate the rapid development of additional chromatid paints, with the goal of ultimately covering the entire genome. These next generation chromatid paints will be of even higher resolution than currently possible, so they will greatly expand potential biodosimetric applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We are developing and refining a technology that allows retrospective assessment of previous DNA damage within an individual as measured by intrachromosal inversions. These mutations are highly stable in the somatic cells of an individual and can serve as biomarkers of previous double stranded DNA breaks long after the damage has occurred. Beyond monitoring DNA damage in astronaut's blood after space missions, the technology will be useful in professions where personnel may be exposed to radiation or clastogenic agents. The technology will also allow discovery of inversions that are associated with human diseases such as cancer and autism.
Task Progress:	Introduction: Kromatid Inc. was awarded a Small Business Innovative Research award from NASA; NASA SBIR Phase 1 (NNX09CE42P) in January 2009, followed by Phase 2 (NNX10CB05C) in January 2010. The project was completed in July 2011. The goal of the SBIR was to develop a method to measure the production of intrachromosomal rearrangements called chromosome inversions and to apply the methodology as another means to assess past exposure to radiation. Before we began, chromosome inversions were not easily detected by existing techniques and small inversions were virtually never detected. Furthermore intrachromosomal rearrangements had been postulated to be a better biomarker of past exposures to relatively low doses of ionizing radiation than other interchromosomal rearrangement, i.e. translocations. Because of the studies of nuclear organization of chromosomal DNA, it was hypothesized that inversions would be particularly indicative of previous exposure to Cosmic Galactic Radiation a type of high energy (LET) radiation that is encountered as astronauts venture further from the protective effect of earth's gravitational field. NASA's interest lies in improving the detection of misrepaired, radiation-induced DNA damage and correlating it to the increased risk of cancer later in life. We completed a convincing prototype of the methodology, i.e. a fluorescent chromatid paint to human chromosome three, and then demonstrated the predicted capability to detect inversions in this chromosome. Inversions appeared as fluorescent signal switches that appeared on the alternate side of chromosome 3 (i.e. the other chromatid) that were easy to score. We then used the assay to count the increased frequency of inversions that were produced after gamma radiation exposure, a proof of concept that the technology could be used for biodosimetry. Since the SBIR ended KromaTiD has progressed toward commercialization of the chromatid paint product and specialty diagnostic probes, recently launching a number of products into the research diagnostic market in the U.S. and Internationally. Relevant to this report and to our goal of developing a biodosimetry product, we collaborated with Dr. Cucinotta's group at NASA and performed four experiments whereby we exposed human lymphocytes ex vivo to three different HZE ions at the NASA NSRL facility at the Brookhaven National Laboratory. The results were very encouraging. The ions, Silicon at 170 MeV/nu, Oxygen at 78 MeV/nu, and Iron at 600 MeV/nu all had more chromosome 3 inversions than two strains of human fibroblasts irradiated with a 6 fold higher dose. These experiments support the hypothesis that HZE ions are more effective at creating intrachromosomal dsDNA breaks that are misrepaired as inversions. Challenge to be addressed by this grant: During the course of the SBIR research, we made 195 fluorescent spots, each spot consisting of 90 oligonucleotides (over 17,000 total) that when labeled with fluorophore, and hybridized to human metaphase spreads produced the first human chromatid paint for the detection of inversions. Chromosome 3 holds 6.6% of the DNA from the human genome. The oligonucleotides were designed to unique DNA sequences (meaning that they only hybridize to one location on any of the 24 human chromosome types under the conditions that we use) and each fluorescent spot making up the paint was spaced approximately 1 million base pairs apart along the length of the chromosome (excluding the repetitive sequences at the centromere). On a genome scale only about one tenth of a percentage point of the genome was covered with oligos. Therefore the probes covering only a small percentage of the genomic sequences produced a reasonable dose response with gamma rays and a robust dose response with HZE ions. It is therefore rational that if more chromatid paints are made and if the number of fluorescent signals on each chromosome is increased, the number of radiation-induced inversions at a given dose that are detected will also increase. A corollary is that if the dose response is increased then the sensitivity of the assay will increase, giving higher statistical significance to lower exposure levels. The problem that we are addressing here is that the synthesis of individual oligonucleotides and subsequent pooling and labeling is expensive and relatively inefficient. While we learned a great deal and made our production more efficient as we progressed through the SBIR, the process of ordering and assembling individual oligonucleotides remains a challenge, especially for producing whole paints for discovery of new recurrent inversions and biodosimetry applications. The goal of the new project begun in September of 2012 is to greatly increase the efficiency of fluorescent probe production by production en masse and then to use this new approach to make additional chromatid paints for biodosimetry. Our ultimate goal is to cover all of the human chromosomes with as many fluorescent signals as possible and to automate the scoring and analyses of inversions. We postulate that by doing this we will be able to increase sensitivity of a biodosimetry assay by a factor of 100 and decrease the associated production costs by a factor of 20. Summary of Approach from the Proposal. The work that was proposed focused on developing a new approach to synthesizing oligonucleotides using a new massively parallel approach meaning the oligos would be synthesized in pools of 4000 oligos at a time. We planned to test the feasibility of this approach by direct comparison to our existing chromosome 3-specific chromatid paint, then if successful, synthesizing the four largest chromosome paints, for chromosomes 1 through 4. A dose response curve was to be generated, the expectation being that the sensitivity of the bioassay will increase proportionally to the increased genome coverage. We hypothesized that the critical experiment that we needed in order to assess the feasibility of the proposed research was whether or not the oligonucleotides that were synthesized using the massively parallel approach could be labeled and used as a chromatid paint with minimal manipulations and whether or not the concentration that was provided

	would allow enough hybridizations to be economically feasible. The answer was no and no. The oligonucleotides could not be labeled and used for hybridization without purification and the amount provided was not sufficient to purify and then use them for multiple hybridization experiments. We estimated that after purification, at best we would be able to do one experiment. Therefore an amplification scheme was designed and tested. The scheme is complex because our probes are all single-stranded and oriented in the same 5' to 3' orientation. This scheme requires an initial dsDNA amplification step followed by a linear amplification step with proprietary steps in between. Using this process we have created a usable 'mini-paint' that provides a chromatid signal comparable but not quite as large as the individual oligo based paint that we created previously. Based on these results we believe that we will be able to complete at least a partial chromosome 1 chromatid paint in the remaining time. Ideally a second year of funding would allow us to complete the ambitious aims as originally proposed.
Bibliography Type:	Description: (Last Updated: 01/11/2021)
Abstracts for Journals and Proceedings	Ray FA, Zimmerman E, Cornforth MN, Bedford JS, Goodwin EH, Bailey SM. "Progress towards a retrospective biodosimetry assay based on chromosomal inversion detection." EPRBioDose 2013. The Joint International Symposium on EPR Dosimetry and Dating and the International Conference on Biological Dosimetry, Leiden, The Netherlands, 24-28 March 2013. Programme and Abstract Book, EPRBioDose 2013. The Joint International Symposium on EPR Dosimetry and Dating and the International Conference on Biological Dosimetry, Leiden, The Netherlands, 24-28 March 2013. , Mar-2013
Articles in Peer-reviewed Journals	Ray FA, Zimmerman E, Robinson B, Cornforth MN, Bedford JS, Goodwin EH, Bailey SM. "Directional genomic hybridization for chromosomal inversion discovery and detection." Chromosome Research. 2013 Apr;21(2):165-74. http://dx.doi.org/10.1007/s10577-013-9345-0 ; PubMed PMID: 23572395 , Apr-2013